

[INTERNAL]

Celery Background Jobs — Design & Operational Guide

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Last Updated: 2026-01-30
Classification: internal

NovaTech's Celery fleet processes document ingestion, embedding generation, reindex triggers, and async notification pipelines. This guide covers broker configuration, task routing, retry strategies, Dead Letter Queue handling, and the Project Phoenix migration to Celery Beat for scheduled ingestion jobs.

1. Broker & Backend Configuration

Celery uses Redis (ElastiCache) as both broker and result backend. Task messages are published to *task* exchange with per-queue routing keys. The result backend uses Redis with a 24-hour TTL for task state storage.

Queue	Concurrency	Priority	Workers (ECS Tasks)
ingest	4	High	8
embed	2	High	12
rerank	8	Medium	4
notify	16	Low	2
dlq	1	Low	1

2. Retry & DLQ Strategy

All tasks use exponential backoff: `max_retries=5`, `countdown=2**attempt*10` (10s, 20s, 40s, 80s, 160s). After exhausting retries, the task serialises its args and kwargs to the **dlq** queue with an `error_reason` field. A Celery Beat job runs daily at 02:00 UTC to report DLQ depth to the `#eng-alerts` Slack channel.

- Embed tasks: idempotent; safe to retry with same `chunk_id`.
- Ingest tasks: use a distributed lock (Redis SET NX) to prevent double-ingest.
- Notify tasks: at-most-once delivery; no retry on 4xx errors from notification providers.

3. Project Phoenix Integration

Project Phoenix (Q3 2026) consolidates four separate cron-based ingestion scripts into a unified Celery Beat schedule. The new schedule includes: hourly Confluence crawl, 6-hourly PDF corpus scan, daily Stack Overflow data refresh, and weekly Qdrant index optimisation. This eliminates 1 200+ lines of bespoke scheduler code and provides a single pane of glass in Flower.

4. Observability

Flower is deployed as an internal-only service (no public ingress) on port 5555. Each task emits structured logs (task_id, queue, attempt, duration_ms) to CloudWatch. A custom Grafana dashboard tracks task throughput, failure rate, and P95 queue wait time per queue. SLA: embed queue P95 wait < 30 seconds.